4.5.2 Chemical cells and fuel cells (chemistry only)

4.5.2.1 Cells and batteries

Content	Key opportunities for skills development
Cells contain chemicals which react to produce electricity. The voltage produced by a cell is dependent upon a number of factors including the type of electrode and electrolyte. A simple cell can be made by connecting two different metals in contact with an electrolyte. Batteries consist of two or more cells connected together in series to provide a greater voltage. In non-rechargeable cells and batteries the chemical reactions stop when one of the reactants has been used up. Alkaline batteries are non-rechargeable. Rechargeable cells and batteries can be recharged because the chemical reactions are reversed when an external electrical current is supplied. Students should be able to interpret data for relative reactivity of different metals and evaluate the use of cells. Students do not need to know details of cells and batteries other than those specified.	AT6 Safe and careful use of liquids.

4.5.2.2 Fuel cells

Content	Key opportunities for skills development
Fuel cells are supplied by an external source of fuel (eg hydrogen) and oxygen or air. The fuel is oxidised electrochemically within the fuel cell to produce a potential difference. The overall reaction in a hydrogen fuel cell involves the oxidation of hydrogen to produce water. Hydrogen fuel cells offer a potential alternative to rechargeable cells and batteries. Students should be able to: • evaluate the use of hydrogen fuel cells in comparison with rechargeable cells and batteries • (HT only) write the half equations for the electrode reactions in the hydrogen fuel cell.	